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CONTROL DEVICE FOR A PUSH BAR LATCH

Abstract

A control device for a push bar latch, comprising: a base frame, the base frame has a pair of horizontal slide grooves and a pair of inclined slide grooves; a control connecting rod; a first positioning pin is arranged at the inner end thereof, the first positioning pin is set through the pair of horizontal slide grooves, and a second positioning pin is set through the pair of inclined slide grooves at the outer end thereof; a triangular blocking member; a locking lever, has a pivot hole pivot in the base frame by a shaft, the locking rod forms a lever structure with the pivot hole as the axis, make the electromagnet be energized to attract the locking rod with a very small suction force to achieve the effect of saving electricity.

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Background/Summary

[0001] This patent application is a continuation-in-part of Ser. No. 18/355,113, filed on Jul. 19, 2023.

BACKGROUND OF THE INVENTION

1. Field of the Invention:

[0002] The invention relates to a control device for a push bar latch, especially one that has an enter control function.

2. Description of the Related Art

[0003] As shown in FIG. 2, the push bar latch **10** can be unlocked by pushing the push rod **11** to go out, and the setting range of the push rod **11** is very large, so that the push bar latch **10** is very convenient for unlocking and is often used in exit doors. However, there are more and more demands for access control, as long as there are no security concerns, adding a control device to the push bar latch **10** can facilitate the entry and exit of people and has an enter control function.

[0004] Referring to FIG. 1, U.S. Pat. No. 5,823,582, for an exit bar for securing a door, discloses a main housing **12'**, a main lock frame **16**, a master main link **32'**, a slave main link **34**, a drive link **62**, and a min spring carrier **110**. The master main link **32'** and slave main link **34** are pivotally connected to a push pad **30'** by pins **36** and **38**. The drive link **62** is connected to the master main link **32'** by link pin **48**. The device also includes an armature **70** and a master auxiliary link **78**. The master auxiliary link **78** is pivotally connected to the drive link **62** by pin **80** and is also connected to an auxiliary armature tray **74** by pin **82**. The device also has a retraction pad **56** and an electromagnet **64**. The electromagnet **64** is used for holding the armature **70** to maintain the blocking member **56** in a locking state by maintaining the blocking member **56** away from the latch retraction lever, and the door is locked.

[0005] However, referring to the structure of U.S. Pat. No. 5,823,582, an electromagnet **64** is arranged at the middle, the armature **70** is arranged above the electromagnet **64**, and the overall mechanical transmission method for unlocking is too complicated, which increases the power consumption of the electromagnet **64**. For example, in order to prevent the door from being pushed open by others, the door lock must have a suction force of at least 500 pounds. However, the electromagnet **64** with a suction force of 500 pounds is very large and takes up a lot of space, which is not conducive to installation, and it consumes a lot of power. Additionally, all of the pins discussed are free floating without being guided by stabilizing slots. Therefore, it requires improvement.

SUMMARY OF THE INVENTION

[0006] It is a primary objective of the present invention to provide a control device for a push bar latch, especially one that has the advantages of simple structure, high control stability, power saving and enter control function.

[0007] In order to achieve the above objectives, an embodiment of the present invention comprises:

[0008] a base frame **20**, a control connecting rod **30**, a locking lever **40**, and an electromagnet **50**, wherein [0009] the base frame **20** has a pair of horizontal slide grooves **21** on both sides of the middle, and a pair of inclined slide grooves **22** are provided in front of the horizontal slide grooves **21**; [0010] the control connecting rod **30** is arranged on the base frame **20**, a first positioning pin **31** is arranged at the inner end thereof, the first positioning pin **31** is set through the pair of horizontal slide grooves **21**, and a second positioning pin **32** is set through the pair of inclined slide grooves **22** at the outer end thereof; furthermore, a torsion spring **13** is arranged at the bottom of the second positioning pin **32**, so that the control connecting rod **30** can be displaced in the pair of inclined

slide grooves **22** and the pair of horizontal slide grooves **21**; [0011] the locking lever **40**, has a pivot hole **42** in the middle thereof and is pivot in the base frame **20** by a shaft **45**, the locking lever **40** has a triangular blocking member **41** at the bottom of the outer end of the pivot hole **42** opposite to the first positioning pin **31**, and a push spring **60** is arranged above the triangular blocking member **41**, a longer inner end section **43** is formed at the other end of the triangular blocking member **41**, which is served as a magnetic attraction surface **431**, so that the locking lever **40** takes the pivot hole **42** as the axis to form a lever structure; and the electromagnet **50** is arranged on the base frame **20** relative to the magnetic attraction surface **431** of the inner end section **43** of the locking lever **40**, and is used for holding the magnetic attraction surface **431** of the inner end section **43** of the locking lever **40**, so let the triangular blocking member **41** enter a locking state for the first positioning pin **31**; whereby, when the electromagnet **50** is energized to magnetize the magnetic attraction surface **431** on the inner end section of the locking lever **40**, letting a front end surface **411** of the triangular blocking member **41** able to block the first positioning pin **31** downward, and enters a locked state (lock), so that the control connecting rod **30** enters a locked state, and the horizontal push bar latch **10** enters a close mode, at this time, even horizontally pushing the push rod **11** cannot unlock the door; Furthermore, when the electromagnet **50** is powered off, it no longer attracts the inner end section **43** of the locking lever **40**, that is, the front end surface **411** of the triangular blocking member **41** releases the blocking state of the first positioning pin **31** and unlocks the door, at this time, the control connecting rod **30** is enable to displace in the pair of inclined slide grooves **22** and the pair of horizontal slide grooves **21**, and the push bar latch **10** enters the open mode (unlock) that returns to the normal use state of the push bar latch **10**, when someone horizontally pushes the push rod **11** at this time, the first positioning pin **31** can pass through the triangular blocking member **41** to unlock the door.

[0012] Also, the push spring is arranged on the base frame by a screw, and a concave hole is arranged on the top surface of the locking lever relative to the push spring.

[0013] Also, the first positioning pin **31** is sleeved with a sliding tube.

[0014] Also, the torsion spring **13** is set on the base frame **20** by a third positioning pin.

[0015] The effects of the present invention include: by the structural design and spatial arrangement of the control connecting rod **30**, the locking rod **40**, and the electromagnet **50**, the locking rod **40** forms a lever structure with the pivot hole **42** as the axis, make the electromagnet **50** be energized to attract the locking rod **40** with a very small suction force to achieve the effect of saving electricity. Therefore, the overall structure is very different from that of U.S. Pat. No. 5,823,582. The present invention cleverly uses the lever principle of mechanics to make the locking rod **40** and the control connecting rod **30** transmit smoothly, and has a simple structure, high control stability and improved power saving.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. **1** is a schematic diagram illustrating the structure of the prior art;

[0017] FIG. **2** is a schematic diagram illustrating the structure of the push bar latch installed with a preferred embodiment of the present invention;

[0018] FIG. **3** is an exploded perspective view of the push bar latch of a preferred embodiment of the present invention;

[0019] FIG. **4** is an exploded perspective view of a preferred embodiment of the present invention;

[0020] FIG. **5** is an assembly perspective view of the main structure of a preferred embodiment of the present invention;

[0021] FIG. **6A** is a sectional view of a preferred embodiment of the present invention in closed mode;

[0022] FIG. 6B is an enlarged view of 6B in FIG. 6A;

[0023] FIG. 7A is a sectional view of a preferred embodiment of the present invention in open mode;

[0024] FIG. 7B is an enlarged view of 7B in FIG. 7A;

[0025] FIG. 8A is a sectional view of a preferred embodiment of the present invention in unlocking mode; and

[0026] FIG. 8B is an enlarged view of 8B in FIG. 8A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0027] Referring to FIGS. 2-8, a push bar latch **10** includes a main body **12** and a push rod **11** arranged on the main body **12**, and a control device **70** is arranged between the push rod **11** and the main body **12**. The control device **70** includes: a base frame **20**, a control connecting rod **30**, a locking lever **40**, and an electromagnet **50**. Certain aspects of the above structure belong to the Prior Art of the horizontal push door lock, and so will not be described in detail here.

[0028] The illustrated embodiment of the present invention includes a base frame **20**, the base frame **20** having a pair of horizontal slide grooves **21** on both sides of the middle. A pair of inclined slide grooves **22** are provided in front of the horizontal slide grooves **21**. The illustrated embodiment of the present invention also includes a control connecting rod **30**, the control connecting rod **30** being arranged on the base frame **20**. A first positioning pin **31** is arranged at the inner end thereof, the first positioning pin **31** is set through the pair of horizontal slide grooves **21**, and a second positioning pin **32** is set through the pair of inclined slide grooves **22** at the outer end thereof. Furthermore, a torsion spring **13** is arranged at the bottom of the second positioning pin **32**, so that the control connecting rod **30** can be displaced in the pair of inclined slide grooves **22** and the pair of horizontal slide grooves **21**. That is, when the push rod **11** is pushed down to unlock, the second positioning pin **32** will move to the lower right in the figure, and the second positioning pin **32** will return to the locked position due to the upward elastic force of the torsion spring **13**. In this embodiment, a sliding tube **311** is sleeved on the first positioning pin **31** to enable the first positioning pin **31** to slide smoothly. In addition, the torsion spring **13** can be mounted on the base frame **20** via a third positioning pin **14**.

[0029] A locking lever **40** has a pivot hole **42** in the middle thereof and is pivoted in the base frame **20** by a shaft **45**, the locking lever **40** has a triangular blocking member **41** at the bottom of the outer end of the pivot hole **42** opposite to the first positioning pin **31**, and a push spring **60** is arranged above the triangular blocking member **41**. A longer inner end section **43** is formed at the other end of the triangular blocking member **41**, which serves as a magnetic attraction surface **431**, so that the locking lever **40** takes the pivot hole **42** as the axis to form a lever structure.

[0030] The electromagnet **50** is arranged on the base frame **20** relative to the magnetic attraction surface **431** of the inner end section **43** of the locking lever **40**, and is used for holding the magnetic attraction surface **431** of the inner end section **43** of the locking lever **40**, enabling the triangular blocking member **41** to enter a locking state for the first positioning pin **31**.

[0031] With the features disclosed above, as FIGS. 6A and 6B show, when the electromagnet **50** is energized to magnetize the magnetic attraction surface **431** on the inner end section of the locking lever **40**, it enables a front end surface **411** of the triangular blocking member **41** to block the first positioning pin **31** downward and enter a locked state (lock), so that the control connecting rod **30** enters a locked state, and the horizontal push bar latch **10** enters a close mode. At this time, even horizontally pushing the push rod **11** cannot unlock the door.

[0032] The push spring **60** of the present invention allows the magnetic attraction surface **431** of the inner end section of the locking lever **40** to remain in contact with the electromagnet **50**, ensuring that the electromagnet **50** can generate an adsorption force on the inner end section of the locking lever **40** when energized. In this embodiment, the push spring **60** is arranged on the base frame **20** by a screw **61**, and a concave hole **44** is arranged on the top surface of the locking lever **40** relative to the push spring **60**, so that the bottom of the push spring **60** is inserted into position.

[0033] Furthermore, referring to FIGS. 7A and 7B, when the electromagnet 50 is powered off, it no longer attracts the inner end section 43 of the locking lever 40. That is, the front end surface 411 of the triangular blocking member 41 releases the blocking state of the first positioning pin 31 and unlocks the door; at this time, the control connecting rod 30 is enabled to displace in the pair of inclined slide grooves 22 and the pair of horizontal slide grooves 21. Then, the first positioning pin 31 passes the triangular blocking member 41, referring to FIGS. 8A and 8B, the push bar latch 10 enters the open mode (unlock) that returns to the normal use state of the push bar latch 10. When someone horizontally pushes the push rod 11 at this time, the first positioning pin 31 can pass through the triangular blocking member 41 to unlock the door.

[0034] Also, the base frame 20 can be a base frame owned by the push bar latch, and the design of the push bar latch can vary. During the unlocking process, in addition to the above-mentioned situation that the second positioning pin 32 can shift to the lower right in the figure, other designs can also be shifted to the lower left in the figure, or even simply shifted to the left or right, and the mechanism of the present invention can be applied to all of them. If the push bar latch is the kind that will make the second positioning pin 32 shift to the right in the figure, the initial position of the first positioning pin 31 will be located on the left side of the triangular blocking member 41, and the electromagnet 50 can hold the triangular blocking member 41 through magnetic force to prevent the first positioning pin 31 from shifting to the right, so that the first positioning pin 31 enters the locked state. If the push bar latch is the kind that will make the second positioning pin 32 shift to the left in the figure, the initial position of the first positioning pin 31 will be located on the right side of the triangular blocking member 41, and the electromagnet 50 can hold the triangular blocking member 41 through magnetic force to prevent the positioning pin 31 from shifting to the left, so that the positioning pin 31 enters the locked state.

[0035] Therefore, by the structural design and spatial arrangement of the control connecting rod 30, the locking rod 40, and the electromagnet 50, the locking rod 40 forms a lever structure with the pivot hole 42 as the axis, enabling the electromagnet 50 to be energized to attract the locking rod 40 with a very small suction force to achieve the effect of saving electricity. For example, in order to prevent the door from being pushed open by others, the door lock must have a suction force of at least 500 pounds. However, the electromagnet with a suction force of 500 pounds is very large and takes up a lot of space, which is not conducive to installation, and it consumes a lot of power. Therefore, it requires improvement. In contrast, the present invention utilizes a longer inner end section 43 that can serve as a magnetic attraction surface 431, and the electromagnet 50 only needs 100 pounds of suction force to attract the locking lever 40, so the electromagnet 50 is small in size, easy to install, and very energy-saving. Therefore, the overall structure of the present invention is very different from that of U.S. Pat. No. 5,823,582. The present invention cleverly uses the lever principle of mechanics to make the locking rod 40 and the control connecting rod 30 transmit smoothly, and has a simple structure, high control stability, and improved power saving.

Claims

1. A control device for a push bar latch, comprising: a base frame (20), a control connecting rod (30), a locking lever (40), and an electromagnet (50), wherein the base frame (20) has a pair of horizontal slide grooves (21) on both sides of a middle section of the base frame (20), and a pair of inclined slide grooves (22) are provided on both sides of the base frame (20) displaced from the horizontal slide grooves (21); the control connecting rod (30) is arranged on the base frame (20), a first positioning pin (31) is arranged at a first end of the control connecting rod (30), the first positioning pin (31) is set through the pair of horizontal slide grooves (21), and a second positioning pin (32) is set through the pair of inclined slide grooves (22) at a second end of the control connecting rod (30); furthermore, a torsion spring (13) is arranged at the bottom of the second positioning pin (32), so that the control connecting rod 30 can be displaced in the pair of

inclined slide grooves (22) and the pair of horizontal slide grooves (21); the locking lever (40), has a pivot hole (42) in a middle section thereof and is pivoted in the base frame (20) about a shaft (45), the locking lever 40 has a triangular blocking member (41) at the bottom of a first end of the locking lever (40) opposite to the first positioning pin (31), and a push spring (60) is arranged above the triangular blocking member (41), a longer second end section (43) of the locking lever (40) is formed at the opposite end of the locking lever (40) having the triangular blocking member (41), which is served as a magnetic attraction surface (431), so that the locking lever 40 takes the pivot hole (42) as the axis to form a lever structure; and the electromagnet (50) is arranged on the base frame (20) relative to the magnetic attraction surface (431) of the second end section (43) of the locking lever (40), and is used for holding the magnetic attraction surface (431) of the second end section (43) of the locking lever (40), so let the triangular blocking member (41) enter a locking state for the first positioning pin (31); whereby, when the electromagnet (50) is energized to magnetize the magnetic attraction surface (431) on the second end section of the locking lever (40), a front end surface (411) of the triangular blocking member (41) blocks movement of the first positioning pin (31), and enters a locked state, so that the control connecting rod (30) enters a locked state, and the horizontal push bar latch (10) enters a close mode, when the electromagnet (50) is de-energized, the second end section (43) of the locking lever 40 is released, wherein the front end surface (411) of the triangular blocking member (41) releases the blocking state of the first positioning pin (31), the control connecting rod (30) is enabled for displacement in the pair of inclined slide grooves (22) and the pair of horizontal slide grooves (21), and the push bar latch (10) enters an open mode that returns to the normal use state of the push bar latch (10), when the push rod (11) is horizontally pushed, the first positioning pin (31) deflects the triangular blocking member (41).

2. The control device for a push bar latch as claimed in claim 1, wherein the push spring is arranged on the base frame by a screw, and a concave hole is arranged on the top surface of the locking lever relative to the push spring.
 3. The control device for a push bar latch as claimed in claim 1, wherein the first positioning pin (31) is sleeved with a sliding tube.
 4. The control device for a push bar latch as claimed in claim 1, wherein the torsion spring (13) is set on the base frame (20) by a third positioning pin.
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